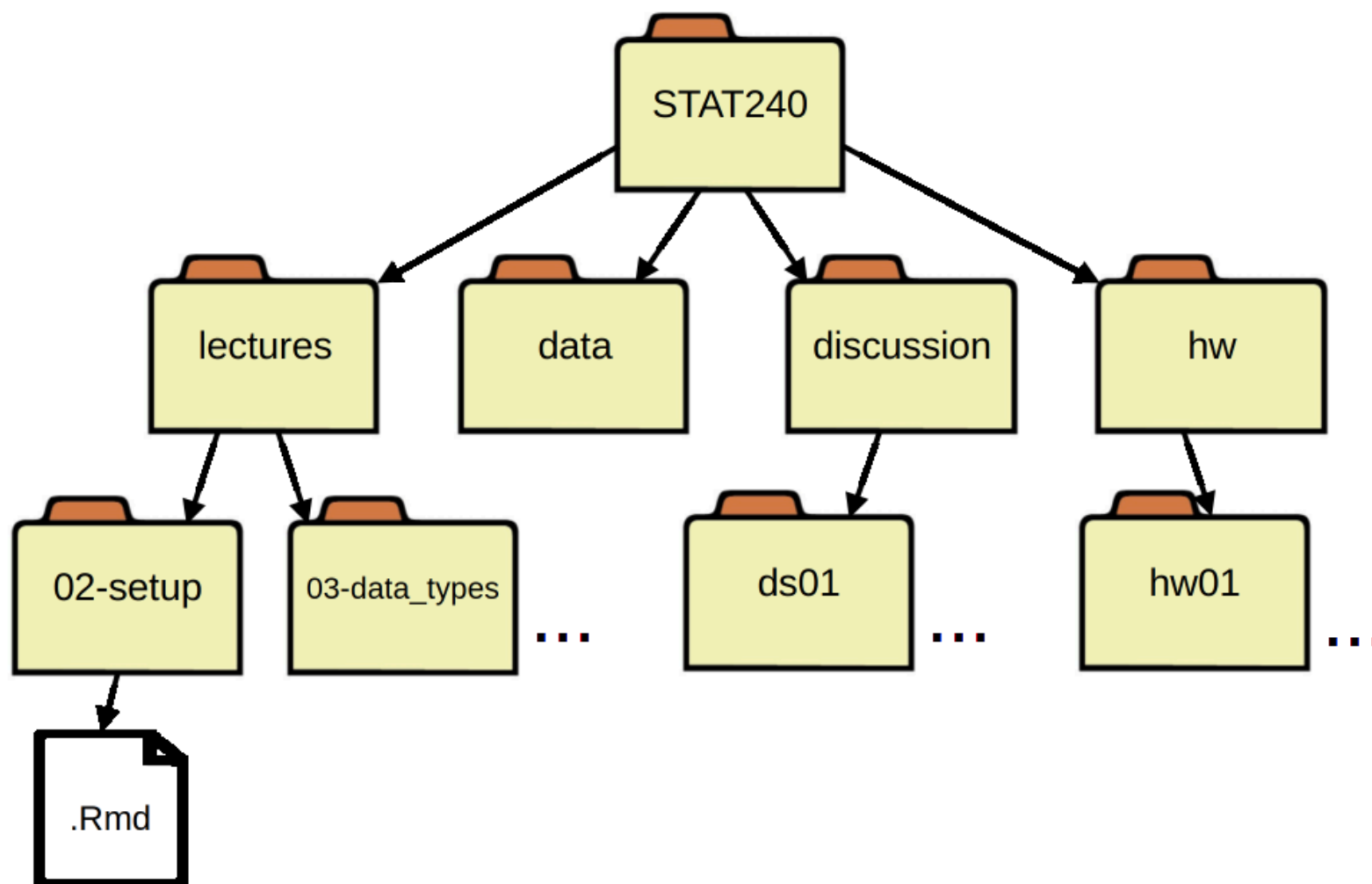


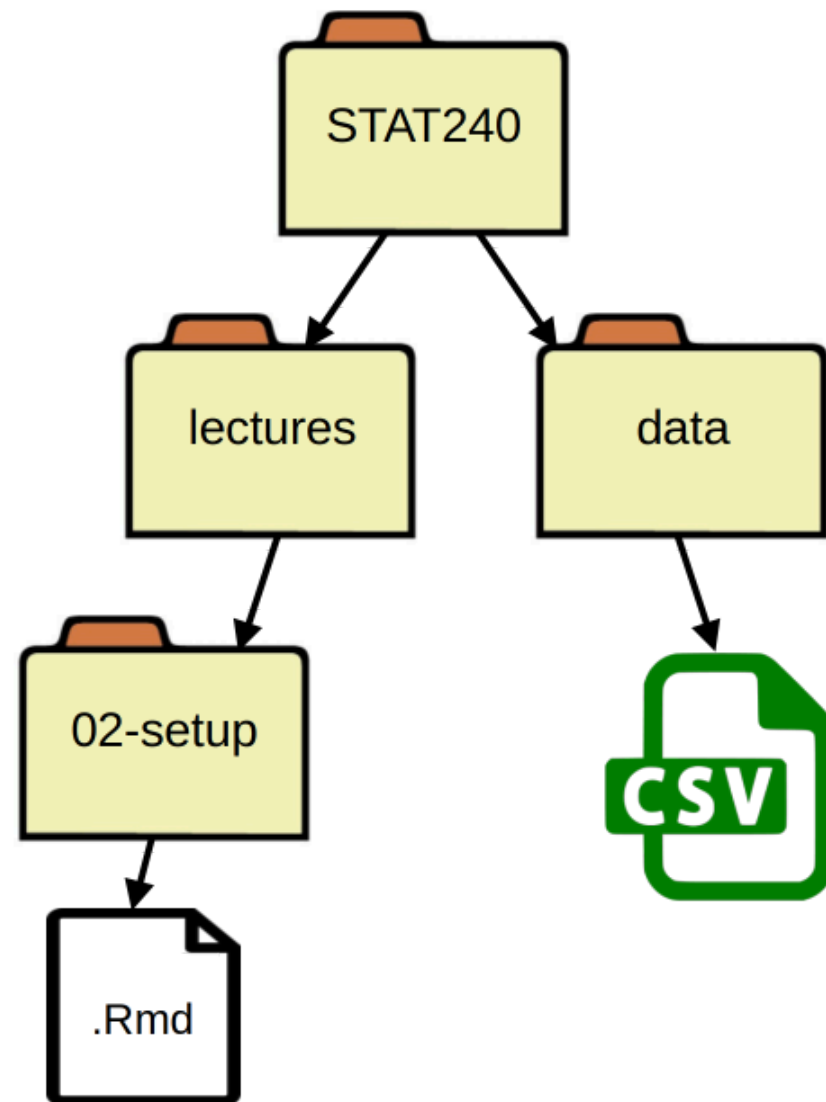
Class Setup (Solutions)

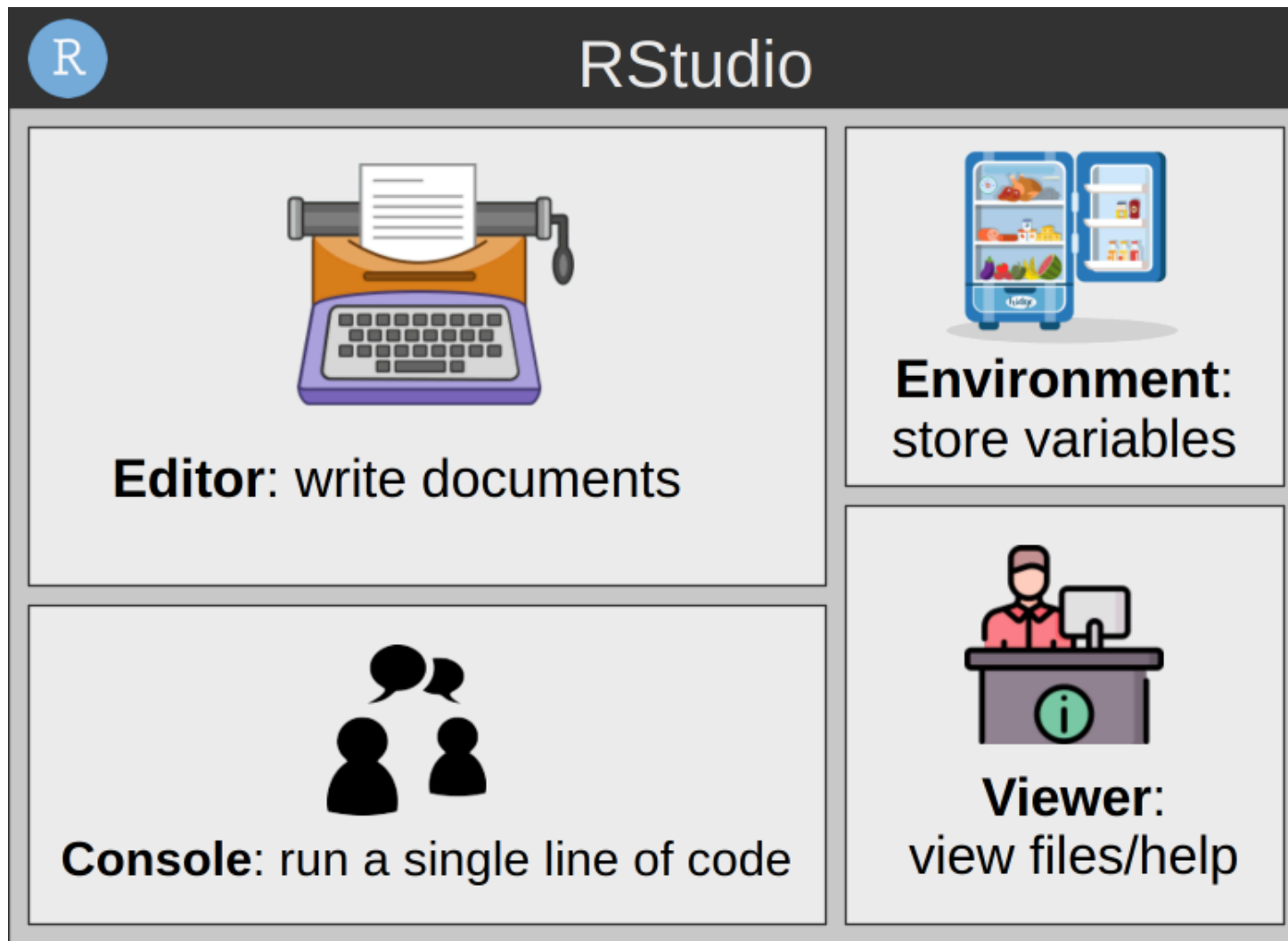
Organization + Intro to R

All of the files (handouts, homework, data, etc.) for 240 will live in folders organized like so:



```
1 df <- read_csv("../..data/example.csv")
```







R/RStudio questions always welcome at office hours or GLC!

Exercises

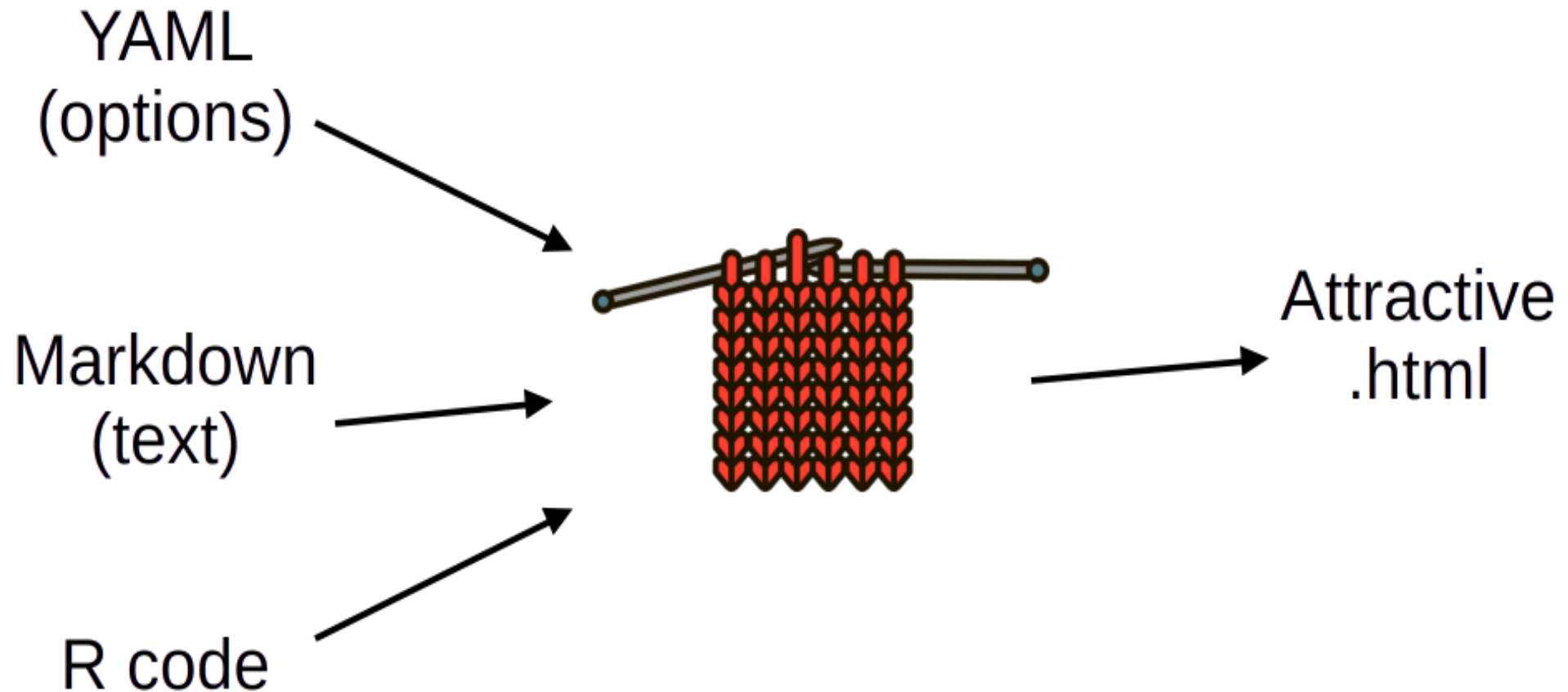
- Perform a mathematical operation with variables (not just numbers)
- Clear your console output - can you still get the old code back?
- Find the Appearance Options menu and change the color scheme
- Figure out how to find your R version (it's a pre-defined object)



Solutions/Hints

- Use `<-` to define a variable
- Find previous lines by pressing “up” or in History
- Tools -> Global Options -> Appearance
- R version is given at the top of console or in an object called `version`

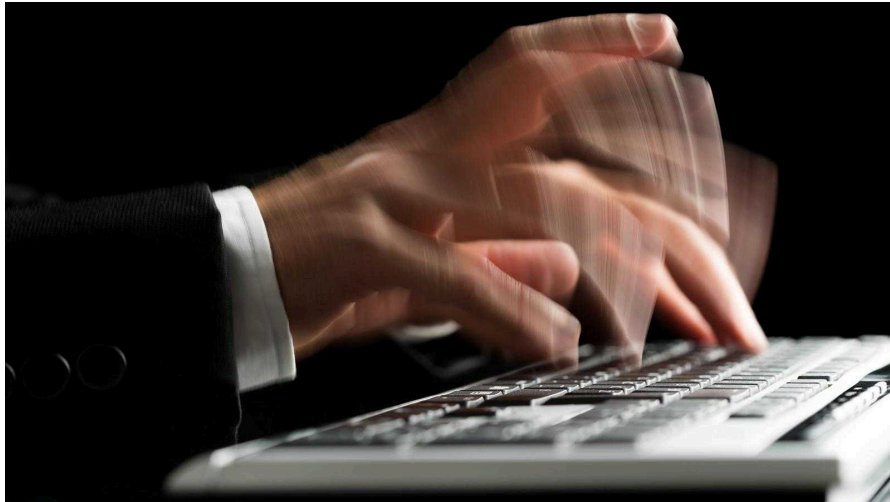
Homework and code handouts have the **R Markdown** format.



Write both [text](#) and R code in the **Environment**, then knit.

Code chunk shortcuts:

	Windows	Mac
Insert code chunk	Ctrl+Alt+I	Cmd+Opt+I
Run a single line of code	Ctrl+Enter	Cmd+Enter



Get used to these!!

Exercises

- Add the line `months <- age * 12` to your Rmd.
- What happens when you run the chunk?
- What happens when you knit?
- Run the line `age <- 20` in your **console**.
- What happens when you knit?



Solutions/Hints

- `months <- age * 12` attempts to use a variable called `age`.
- Running the chunk should give an error in the console.
- Knitting the file should give the same error.
- After running this line, the chunk should run **correctly** with the green play button.
- Knitting the file should still give the same error as before.

Thursday, 11:30pm



My chunks all run perfectly, but my file won't knit! I'm getting a "object not found" error but it's right there in the Environment.

Knitting a file creates a completely separate R instance - it can't see your Environment tab. Make sure all objects are defined in the .Rmd itself, and don't rely too much on the console!

We'll be using the [tidyverse](#) package, an extra set of software for R. A package is like a toolbox.



Installation is like
buying a toolbox



Loading is like
opening a toolbox

Installing tidyverse

In your console, run

```
1 install.packages("tidyverse")
```

You only need to do this once!

Additional troubleshooting

- `use a personal library`: pick yes
- `install from source`: pick no, then yes if it fails
- `compile`: pick no

Installation could take several minutes.

Looking ahead

In [chapter 3](#), we'll learn the basics of writing code in R. If you want a head start, try to figure out on your own what the following lines of code do:

```
1 favorite_numbers <- c(3, 1, 5, 6)
2 favorite_words <- c("apple", "mystical", "obsequious")
3
4 favorite_numbers[3]
5 favorite_numbers^2
6
7 str(favorite_words)
8 paste(favorite_words, "ly", sep = "")
```